

Text Mining Project: Predicting Song Popularity through Lyrics and Features

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Abstract

This document outlines a text mining project that predicts song popularity, measured by views, using the Genius Song Lyrics dataset. The dataset includes lyrics, titles, features, and artist information, with views as the target variable. Preprocessing steps include song part indicator removal and feature cleaning. Text representation techniques such as TF-IDF, Bag-of-Words, GloVe, and BERT are employed, capturing both surface-level and deep semantic features. Regression models like SVM and DNN are used to predict views, with performance evaluated by accuracy and weighted F1-score. The study highlights the value of combining text-based and metadata features, offering insights into the factors influencing song success.

1 Introduction

As the music industry continues to expand, millions of new songs are released every day across various platforms. This unprecedented volume has created a highly competitive landscape where the ability to predict and understand the factors driving a song's success has become a valuable asset for composers, producers, and record labels alike. In such a dynamic field, success often hinges on the interplay of multiple elements, including lyrical content, genre, artist recognition, and cultural trends. However, many composers and producers today still rely primarily on personal experience and intuition when deciding whether to release a song. While experience has historically been a cornerstone of artistic and commercial success, its limitations have become increasingly evident in today's data-driven environment.

Human judgment, no matter how experienced, is prone to biases, constrained by memory, and lacks the capacity to process the immense volumes of data now available that influence a song's reception. These limitations are particularly pronounced in an industry where digital metrics, such as streaming counts, social media trends, and online reviews,

play a growing role in shaping the perception and popularity of music. As a result, relying solely on subjective evaluation can lead to missed opportunities, ineffective marketing strategies, or misjudged potential, particularly for new and emerging artists seeking to break into the market.

This study aims to address these challenges by providing a framework for researchers to explore the application of text mining in analyzing song lyrics and related features to predict a song's popularity. The methods outlined here are not exhaustive but are rooted in established principles of text mining and classification models. Specifically, this document provides guidance on preprocessing, feature extraction, and algorithm selection to facilitate predictive analysis. The dataset includes key elements such as lyrics, song titles, associated features, artist information, and view counts, the latter serving as the primary target variable. By integrating these elements into predictive models, this study seeks to offer insights into the factors influencing song popularity and to provide tools for data-driven decision-making in the music industry.

2 Related Works

Prior research has frequently explored the use of song lyrics for various classification tasks, such as identifying explicit content in music and predicting genres. For instance, [Egivenia et al. \(2021\)](#) developed a hybrid model that combined word embeddings with recurrent neural networks (RNNs) to detect explicit content in lyrics. Similarly, [Akalp et al. \(2021\)](#) employed convolutional neural networks (CNNs) to identify genre-specific patterns in lyric texts. These studies underscore the potential of lyrics as a rich source of information for various predictive tasks, demonstrating their utility in both content analysis and classification.

Despite this progress, research specifically focused on predicting song popularity using lyrics remains relatively underexplored. In contrast, other

Title	Tag	Artist	Year	Views	Features
<i>Killa Cam</i>	rap	Cam'ron	2004	173,166	{"Cam'ron", "Opera Steve"}
<i>Can I Live</i>	rap	JAY-Z	1996	468,624	{}
<i>Forgive Me Father</i>	pop	Fabulous	2003	4,743	{}

Lyrics	ID	Language (cld3)	Language (ft)
[Chorus: Opera Steve & Cam'ron] Killa Cam, Killa Cam, Cam Killa Cam, Killa Cam Killa Cam, Cam Killa ...	1	en	en
[Produced by Irv Gotti] [Intro] Yeah, hah, yeah, Roc-A-Fella We invite you to somethin' epic, you k...	3	en	en
Maybe cause I'm eatin And these bastards fiend for they grub I carry pumps like I serve gasoline to ...	4	en	en

Table 1: Examples of data points

forms of textual data, such as social media content, have been extensively studied in the context of popularity prediction. For example, [de Sá et al. \(2021\)](#) and [Trzciński and Rokita \(2017\)](#) utilized models such as support vector machines (SVMs), logistic regression, and random forests to predict the popularity of social media posts. Similarly, [Wu et al. \(2022\)](#) integrated visual and textual features to predict engagement on social platforms, demonstrating the value of multimodal approaches in understanding popularity dynamics.

Building on these approaches, this study bridges the gap by focusing specifically on lyrics-based popularity prediction while integrating additional features such as tags, artist information, and collaborations. To achieve this, we designed a comprehensive model that leverages various text representation techniques, including Bag of Words, TF-IDF, GloVe, and BERT. These representations capture both the surface-level and deep semantic features of lyrics, which are then combined with supplementary metadata to provide a more holistic view of the factors influencing song popularity. By employing a range of regression models, such as support vector regression (SVR) and neural networks, this study aims to establish a robust framework for predicting song views, addressing the current gap in lyrics-based prediction research, and advancing the understanding of what makes a song successful.

3 Dataset

To analyze the influence of song lyrics and features on views, we use the Genius Song Lyrics dataset ([GDCJ \(2020\)](#)), which is available on Kaggle. This dataset includes fields such as lyrics, title, features, and artist information. Table 1 displays examples of data points. The views of each song will be our target variable. For this project, we will focus on songs in English only, excluding those in other

languages.

Below are the attributes we focus on:

- **lyrics:** Lyrics
- **title:** Title of the piece. Most entries are songs, but there are also some books, poems and even some other stuff
- **artist:** Person or group the piece is attributed to
- **feature:** Other artists that contributed
- **tags:** Genre of the piece. Most non-music pieces are "misc", but not all. Some songs are also labeled as "misc"
- **views:** Number of page views

This dataset contains 3,374,189 data points, making it ideal for our text mining tasks and serving as the primary data source for our analysis. We use 'views' as the target variable for predicting popularity, while the remaining attributes are utilized to train the models.

4 Methods

4.1 Preprocessing

To prepare the data for text mining, several preprocessing steps are necessary to clean and normalize the song data.

4.1.1 Input

These steps ensure that the text and features are suitable for the representation and predictions applied in the next stages.

- **Song Part Indicator Removal:** All non-lyrical parts of the text, such as "[Intro]",

Title	Tag	Artist	Views	Features	Lyrics
<i>Killa Cam</i>	rap	Cam'ron	Medium	Cam'ron Opera Steve	Killa Cam, Killa Cam, Cam Killa Cam, Killa ...
<i>Can I Live</i>	rap	JAY-Z	Medium		Yeah, hah, yeah, Roc-A-Fella We invite...
<i>Forgive Me Father</i>	pop	Fabulous	Very Low		Maybe cause I'm eatin And these bastards ...

Table 2: Data points after preprocessing

Label	Number
Very Low	680,270
Low	672,183
Medium	675,187
High	672,114
Very High	674,444

Table 3: Number of data points in each bin

"[Chorus]" and "[Verse]," are removed to focus solely on the meaningful content of the song lyrics.

- **Feature Cleaning:** Punctuation and other irrelevant symbols within the feature column (e.g., artist collaborations or special characters) are removed to prevent them from influencing the text representations.

Cleaned data points example are shown in Table 2. The cleaned and preprocessed data will then be used in subsequent steps for generating representations applying predictive models.

4.1.2 Label

Due to the challenges of accurately predicting exact view counts, we first use quantile-based binning to bin the views into categories.

Table 3 shows the number of data points in each bin.

4.2 Song Representation

To predict views, we first generate song representations, which are formed by concatenating tag embeddings, artist embeddings, and song document embeddings. The process for generating these representations is illustrated in Figure 1.

4.2.1 Tag Embedding

The "tag" is treated as a categorical attribute containing 6 unique labels. It is processed through a simple embedding layer to generate 2-dimensional embeddings.

4.2.2 Artists Embedding

The "artist" is treated as a categorical attribute, similar to the "tag," but with 429,060 unique labels. It

is processed through an embedding layer to generate 600-dimensional embeddings. After generating the embeddings, we create a dictionary that maps each artist to their corresponding embedding.

The "features" attribute of a song consists of a list of all collaborating artists. For each artist in this list, we retrieve their embedding from the artist embedding dictionary. If the list is not empty, the features embedding is computed as the average of all artist embeddings in the list. If the list is empty, the features embedding is set to be identical to the artist embedding.

Finally, the song's artists embedding is defined as the average of the artist embedding and the features embedding.

4.2.3 Song Document Embedding

A song document is defined as the concatenation of its "title" and "lyrics." This document is processed using four different encoding methods to generate its representation:

- **TF-IDF:** The document is represented using the TF-IDF matrix.
- **Bag-of-Words:** The document is represented using count-based indicators.
- **GloVe:** The "en_core_web_md" model is used to tokenize the document. The document representation is derived from the average word embedding of all tokens, resulting in a 300-dimensional output for the document embedding.
- **BERT:** We fine-tuned the "bert-base-uncased" model on all the song document we used. The document representation is obtained from the [CLS] embedding, producing a 768-dimensional output for the document embedding.

4.3 Model Training and Prediction

After generating and processing the features, the next step is to train the models and make predictions based on the different representations.

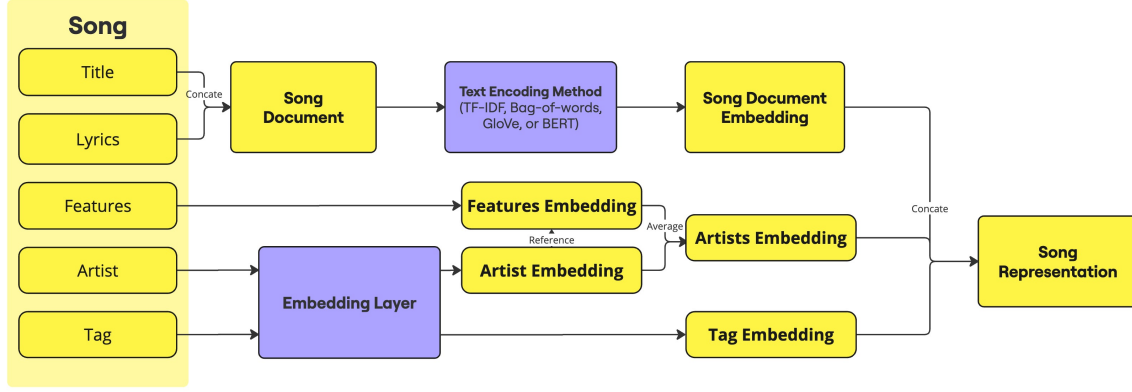


Figure 1: Song representation generation process

- **SVR:** Train the SVR model using the feature vectors derived from the TF-IDF and Bag of Words representations.
- **Neural Network Training:** Use a simple neural network with a fully connected layer to process the embeddings generated from BERT and GloVe, and train it to make predictions.

5 Experiment

Due to the significant time and memory requirements for training on the entire dataset, as well as the lack of access to GPU resources, we limited our training and testing to 100,000 data points, with 20,000 randomly selected from each bin. Although the training and testing data are limited to these 100,000 points, the tag and artist embeddings are generated using the full dataset, as the embedding layers process all the data. This ensures that all tags and artists are represented, even if they are not included in the random sample.

The dataset will be split into training and testing sets with a 9:1 ratio. After training the prediction model, we will evaluate its performance using accuracy and F1-score.

5.1 Ablation of Representations

Before initiating the training of our actual models, we first analyzed the performance impact of each attribute used. Given that SVM is a simpler model compared to neural networks, we conducted an ablation study on this model to assess the performance of various representations. The results are detailed in the Evaluation Matrix presented in Table 5.

In the first set of three models, we did not use any document representations and relied solely on

tag and artist embeddings. The results reveal that while tag embeddings alone yield better accuracy with a score of 0.2426, they underperform in F1-score with only 0.1888 compared to artist embeddings alone. However, when both tag and artist embeddings are applied, the performance improves significantly in both accuracy and F1-score.

For the next four models, which utilize Bag-of-Words (BoW) as the text encoding method, we observed that the model without either tag or artist embeddings performed the worst, with an accuracy of 0.2366 and an F1-score of 0.2361. The model featuring only artist embeddings delivered the best performance among all BoW-based models. Although combining both artist and tag embeddings did not yield the best results among the BoW models, it still surpassed the performance of the model with only tag embeddings, achieving an accuracy of 0.2502 and an F1-score of 0.2455.

For the final four models using TF-IDF as the text encoding method, a similar pattern was observed. Models lacking both tag and artist embeddings performed the poorest, with an accuracy of 0.2588 and an F1-score of 0.2563. The model with only tag embeddings outperformed all other TF-IDF-based models, recording an accuracy of 0.2729 and an F1-score of 0.2698. The combination of both artist and tag embeddings, while not the best, improved performance over the model with only artist embeddings, achieving an accuracy of 0.2706 and an F1-score of 0.2675.

Across all models, it is evident that employing either tag or artist embeddings generally enhances performance, underscoring their potential to improve outcomes. Typically, one type of embedding tends to yield the best results: artist embeddings excel in Bag-of-Words (BoW) models, while

Text Encoding Method	Tag	Artists	Accuracy	F1-score
-	✓		0.2426	0.1888
-		✓	0.2218	0.2202
-	✓	✓	0.2408	<u>0.2408</u>
BoW			0.2366	0.2361
BoW	✓		0.2446	0.2438
BoW		✓	<u>0.2633</u>	<u>0.2608</u>
BoW	✓	✓	0.2502	0.2455
TF-IDF			0.2588	0.2563
TF-IDF	✓		<u>0.2729</u>	<u>0.2698</u>
TF-IDF		✓	0.2633	0.2608
TF-IDF	✓	✓	0.2706	0.2675

Table 4: Evaluation Matrix For Ablation of Representations Study (Single underline denotes the best performance within each text encoding method; double underline indicates the best performance across all models.)

tag embeddings perform better in TF-IDF models. This differentiation may stem from the correlation between artists and tags, as artists often stick to specific music genres, which could diminish the effectiveness of using either tag or artist embeddings alone. Due to our limited computational resources, we could not test all possible embedding combinations, particularly in computationally demanding neural network models. Nevertheless, models incorporating both types of embeddings frequently achieve the second-best performance. Therefore, for our comparative analysis, we decided to train models using both artist and tag embeddings.

5.2 Model Comparison

Using the data described above, we train and evaluate models that utilize document embeddings, with both artist and tag embeddings concatenated.

Table 5 presents the evaluation matrix.

5.2.1 SVM

We trained two LinearSVM models using song representations derived from the TF-IDF and Bag-of-Words methods. The TF-IDF-based model achieved an accuracy and F1-score of 0.27, whereas the Bag-of-Words-based model recorded an accuracy and a weighted F1-score of 0.25.

Comparing the two models, the TF-IDF-based representation outperforms the Bag-of-Words. This superiority is likely due to TF-IDF’s ability to assign greater importance to less common words, which helps capture the distinctive characteristics of a song and enhances its predictive power regarding popularity.

5.2.2 Neural Network Training

We trained two two-layer neural networks, each with a batch size of 32 for 20 epochs, utilizing a validation split of 0.2, to evaluate song representa-

tions derived from GloVe and BERT. Both models were optimized using the Adam optimizer with a learning rate of 0.001 and employed categorical cross-entropy as the loss function.

The GloVe-based model achieved an accuracy of 0.30 and a weighted F1-score of 0.29 on the test set. In contrast, the BERT-based model attained an accuracy of 0.28 and a weighted F1-score of 0.26. Comparing the two models, GloVe clearly outperformed BERT. This superior performance may be attributed to GloVe’s effectiveness in leveraging the statistical significance of word frequencies, which is particularly advantageous in scenarios like song lyrics where word repetition may correlate with popularity. Since GloVe models word significance based on overall word occurrences in a corpus, it effectively captures relationships between words based on their frequency of co-occurrence. This makes it well-suited to capture the trend of repeated words in lyrics. BERT, focusing more on capturing contextual nuances, may overlook the simplistic yet impactful pattern of word frequency, thus underperforming in this specific context.

Comparing all the models above, GloVe performs the best. This suggests that mere occurrences of words may not provide a sufficient representation of lyrics for predicting song popularity. Since the actual content of the lyrics likely contributes to a song’s popularity, the lack of semantic understanding in models like TF-IDF and Bag-of-Words may lead to their underperformance compared to GloVe and BERT-based representations, which incorporate semantic information.

6 Discussion

This study has shed light on the potential of using text mining and machine learning techniques to

Text Encoding Method	Model	Accuracy	F1-score
TF-IDF	SVM	<u>0.27</u>	<u>0.27</u>
BoW	SVM	0.25	0.25
GloVe	NN	<u>0.30</u>	<u>0.29</u>
BERT	NN	0.28	0.26

Table 5: Evaluation Matrix For SVM and Neural Network models (Single underline denotes the best performance within each text encoding method; double underline indicates the best performance across all models.)

predict song popularity by analyzing lyrics and incorporating additional features. Among the models tested, the TF-IDF-based SVM model achieved the highest accuracy, highlighting its ability to emphasize unique and distinctive terms within song lyrics. This characteristic allows it to capture lexical features that are strongly correlated with a song’s popularity, such as rare but impactful words. On the other hand, the GloVe-based neural network model demonstrated the highest F1-score, reflecting its strength in capturing semantic nuances and contextual relationships in the text. These findings suggest that while accuracy and F1-score are interrelated, they represent different dimensions of model performance, each emphasizing distinct aspects of prediction quality.

The inclusion of additional features, such as tags and artist embeddings, further enhanced the performance of all models, with notable improvements observed in the SVM models. This indicates that metadata plays a critical role in predicting song popularity, offering valuable context that complements the analysis of lyrics alone. Tags provide genre-specific information, while artist embeddings encapsulate a wealth of background data about the creators, including their historical popularity and collaborative patterns. By integrating this metadata, the models achieved a more comprehensive understanding of the factors that drive a song’s success.

Despite these achievements, the study faced certain constraints due to computational limitations. The models were trained on a sample of 10,000 data points, which, while sufficient to validate the feasibility of the approach, may not fully capture the complexity and variability of the entire dataset. Training on a larger dataset is expected to enhance the robustness and generalizability of the models. Currently, training was conducted on local personal computers with limited memory capacity, which necessitated this compromise. Moving forward, the transition to cloud-based platforms such as Google Colab will enable the use of greater computational resources, including GPUs and larger RAM, allow-

ing the analysis to scale up significantly.

An additional area of exploration involves testing neural network models without incorporating supplementary features such as tags and artist embeddings. This will help determine the extent to which these features contribute to the observed improvements in performance. By isolating the impact of contextual metadata, the study could uncover whether neural networks can achieve comparable results solely through advanced text representation techniques. Similarly, experimenting with combinations of other features, such as song release year, user interactions, or acoustic properties, may provide new insights into the multifaceted nature of song popularity.

Lastly, while the current approach focuses on predicting the popularity of existing songs, an intriguing extension of this research could involve applying these methods to unreleased tracks. This would allow composers, producers, and record labels to assess the potential reception of their songs before release, aiding in strategic decision-making. Moreover, the integration of these predictions into marketing strategies, such as identifying which songs to promote more heavily or target at specific demographics, could further expand the practical applications of this research.

7 Conclusion

This study demonstrates the value of combining text mining and machine learning techniques with contextual metadata to predict song popularity. The findings highlight the complementary strengths of different models and representations. The TF-IDF-based SVM model excelled in accuracy, likely due to its ability to capture distinctive lexical features, while the GloVe-based neural network achieved the highest F1-score, showcasing its ability to encapsulate semantic and contextual information effectively. Together, these results illustrate that no single approach dominates across all metrics, but rather that the strengths of each method can be leveraged depending on the specific goals of the

analysis.

Incorporating metadata such as tags and artist embeddings proved to be particularly impactful, enriching the models with additional context that lyrics alone cannot provide. This underscores the importance of adopting a holistic approach when analyzing factors that contribute to song popularity. Future research should aim to scale the analysis to the full dataset by leveraging cloud-based resources, such as Google Colab, to enhance computational capacity and enable more comprehensive training. Furthermore, testing additional features and refining existing representations will be crucial to further optimize the models.

Beyond the technical advancements, the practical implications of this research are significant. By providing a data-driven framework for predicting song popularity, this study offers valuable tools for composers and producers in an increasingly competitive music industry. These tools can help stakeholders make informed decisions about song releases, marketing strategies, and resource allocation. With further refinement and validation, this framework has the potential to revolutionize the way music professionals approach the creation and promotion of songs, offering a competitive edge in a data-driven era.

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